MathVerity Prototype Materials

These materials show how learners in grades 5–6, their parents and the internal project team will interact with the planned MathVerity MVP. The key distinction is visible throughout: practice may use hints; independent assessment records a first answer without built-in help. Reports explain the evidence rather than equating XP with mastery.

Material	Purpose	Location in the package
Interactive prototype	Try the main scenarios using synthetic data.	prototype/index.html
Eight English design mockups	Review the key screens and their expected behaviour.	materials/mockups/01 to 08
Two language illustrations	Inspect Kazakh and Russian practice wording.	materials/mockups/09 and 10
Walkthrough and checks	Reproduce scenarios and understand verification limits.	README.md and TESTING.md

Status as of 23 October 2026: an interaction prototype is available through GitHub pages. It does not implement the production React client, Go API, PostgreSQL storage, authentication or a live AI provider. All people and learning records shown in family screens are fictional examples. To interact with prototype, open a URL given in README.md as prototype URL.

Scenario coverage and observable results

Each scenario specifies an actor, a visible action and an expected response. The numbered mockups on the following pages illustrate the intended screens.

Scenario	Actor and entry point	Input or action	Expected response
P01 Start a session	Learner → Overview	Choose one of four topic blocks.	A task and a learning goal open; the next action is clear.
P02 Practise with support	Learner → Practice	Enter an answer; request up to three hints.	Feedback and hint source are visible; assistance is recorded separately.
P03 Work independently	Learner → Independent check	Submit a valid first answer and an explanation choice.	No hint control; the first answer is locked and XP is awarded once.
P04 Understand a result	Learner → Results	Read accuracy, dates, templates and evidence status.	One correct answer still shows insufficient evidence.
P05 Read family progress	Parent → Child overview	Open the prepared example or current walkthrough.	A linked child’s statuses and a review recommendation appear.
P06 Inspect a skill report	Parent → Skill report	Compare eight skills and their evidence.	Review, developing, confirmed, stale and insufficient states are distinct.
P07 Link or revoke access	Parent and learner → Family linking	Generate and enter a code; try invalid or expired codes.	A valid unused code links the profiles; revoked links hide reports.
P08 Review original content	Internal team → Peer review	Request review, check the content, approve and publish.	Self-approval is blocked; the other developer reviews a version.

Learner overview

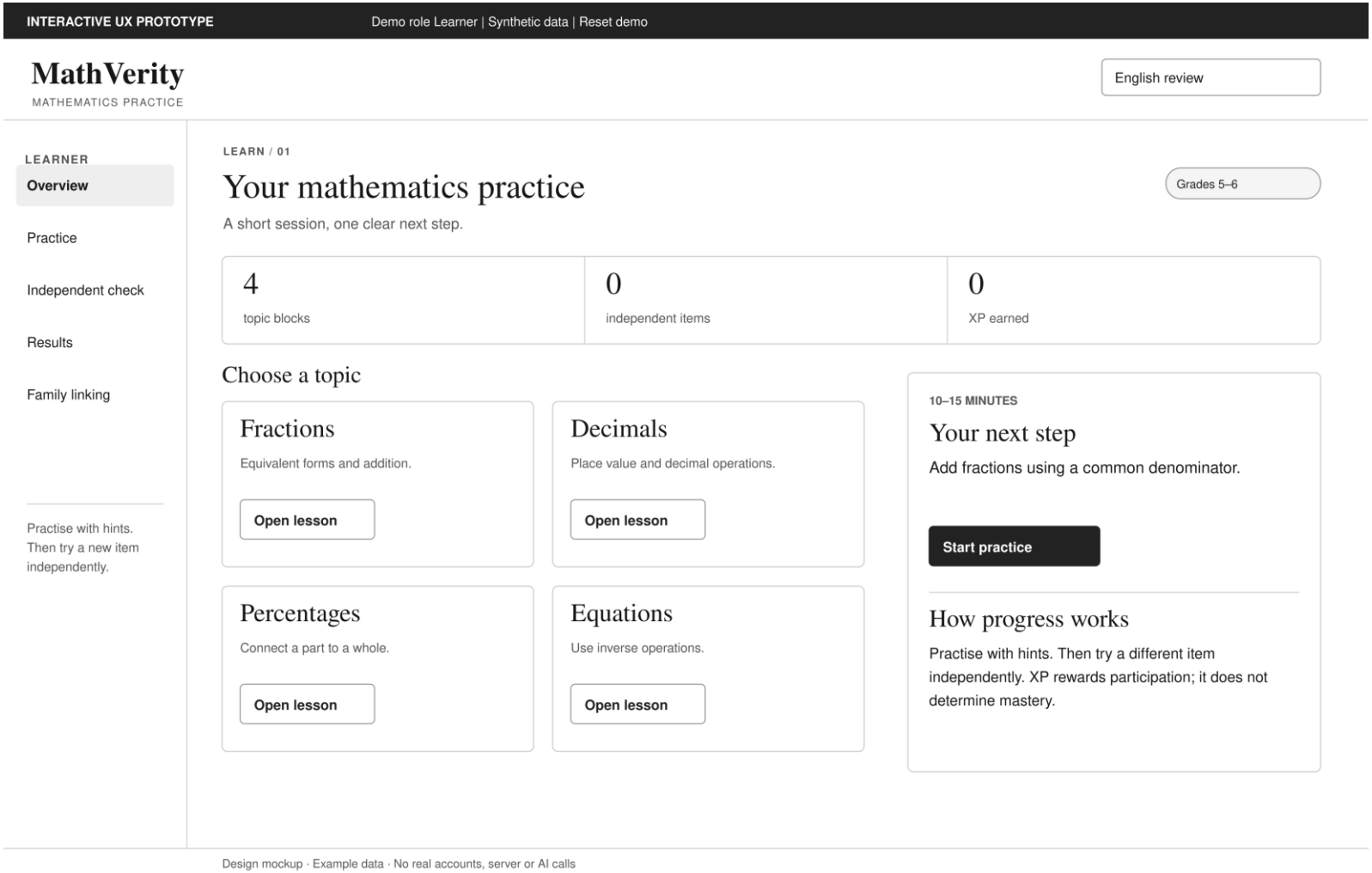


Figure 1. Entry to a short mathematics session.

The learner selects a topic or follows the suggested next step. The interface separates activity and XP from independent evidence.

Assisted practice

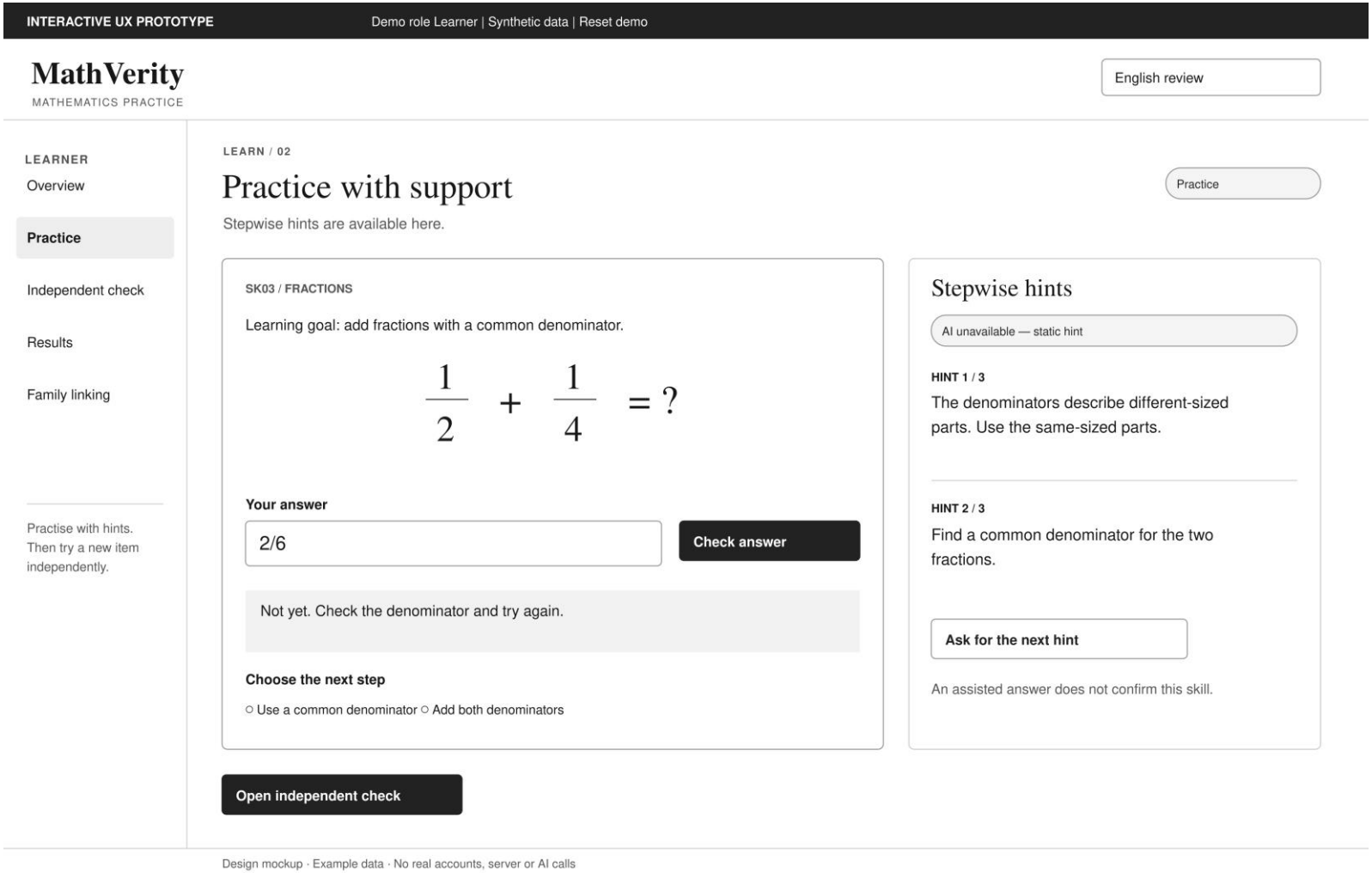


Figure 2. Incorrect practice answer and staged hints.

The learner can retry and request a next hint. The source label shows a static fallback when AI is unavailable; practice does not confirm mastery.

Independent assessment

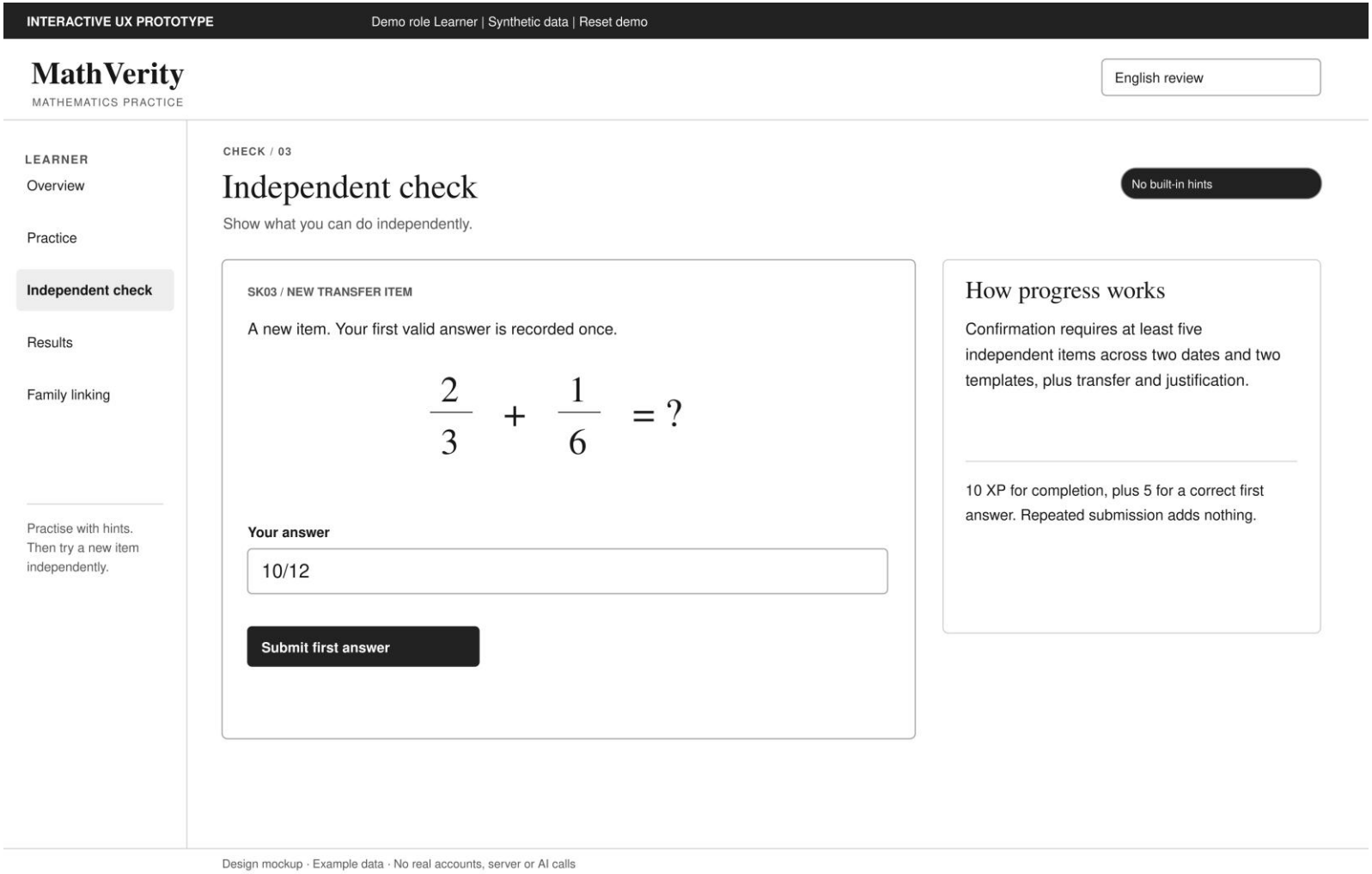


Figure 3. First answer submitted without built-in hints.

The independent item differs from the practice item. A malformed answer is rejected without recording an attempt; the first valid answer is retained.

Results and supporting evidence

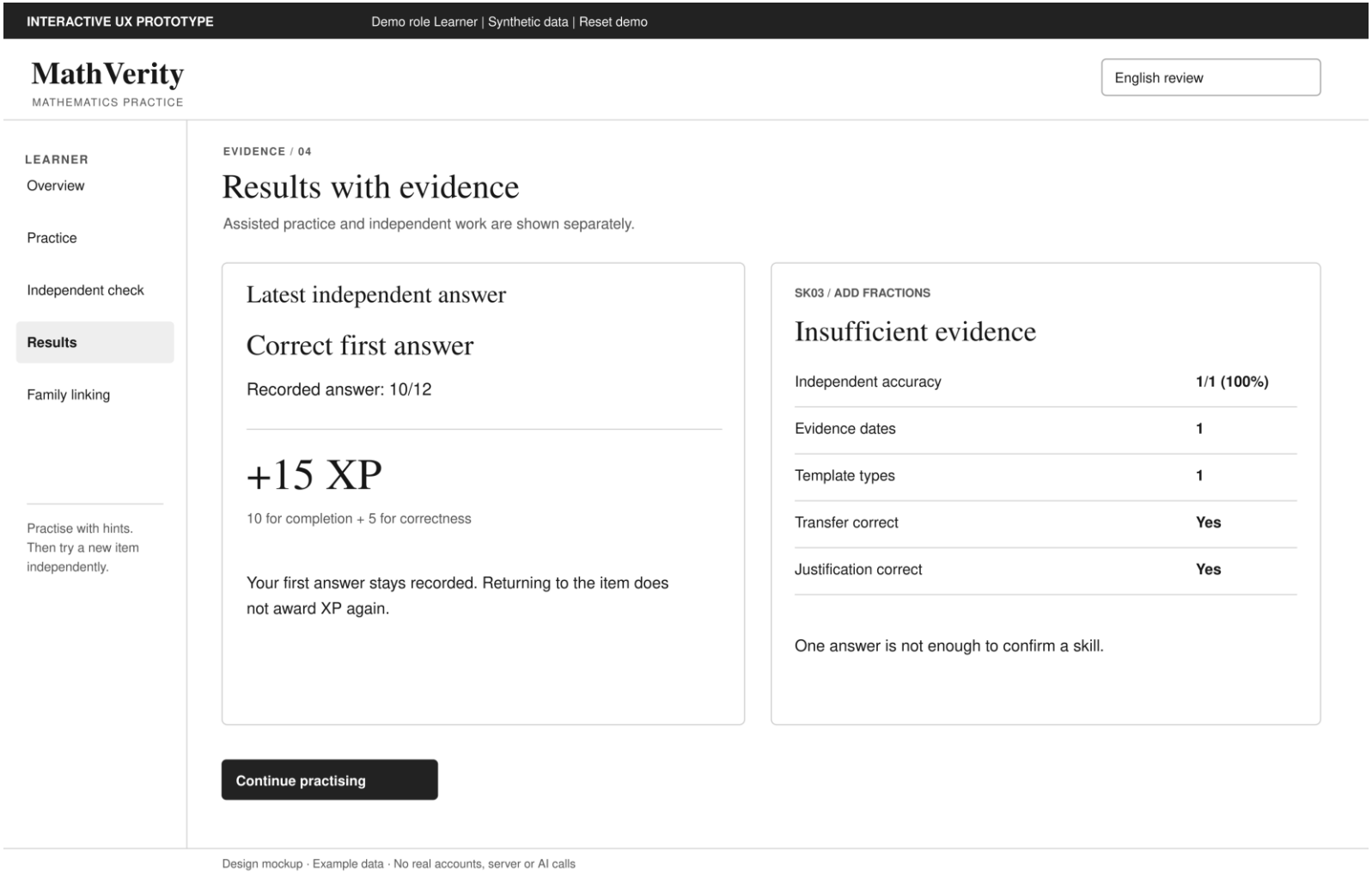


Figure 4. Correct answer with insufficient evidence.

A correct first answer earns 15 XP once. Accuracy of 1/1 is shown alongside missing evidence; it does not confirm a skill.

Parent overview

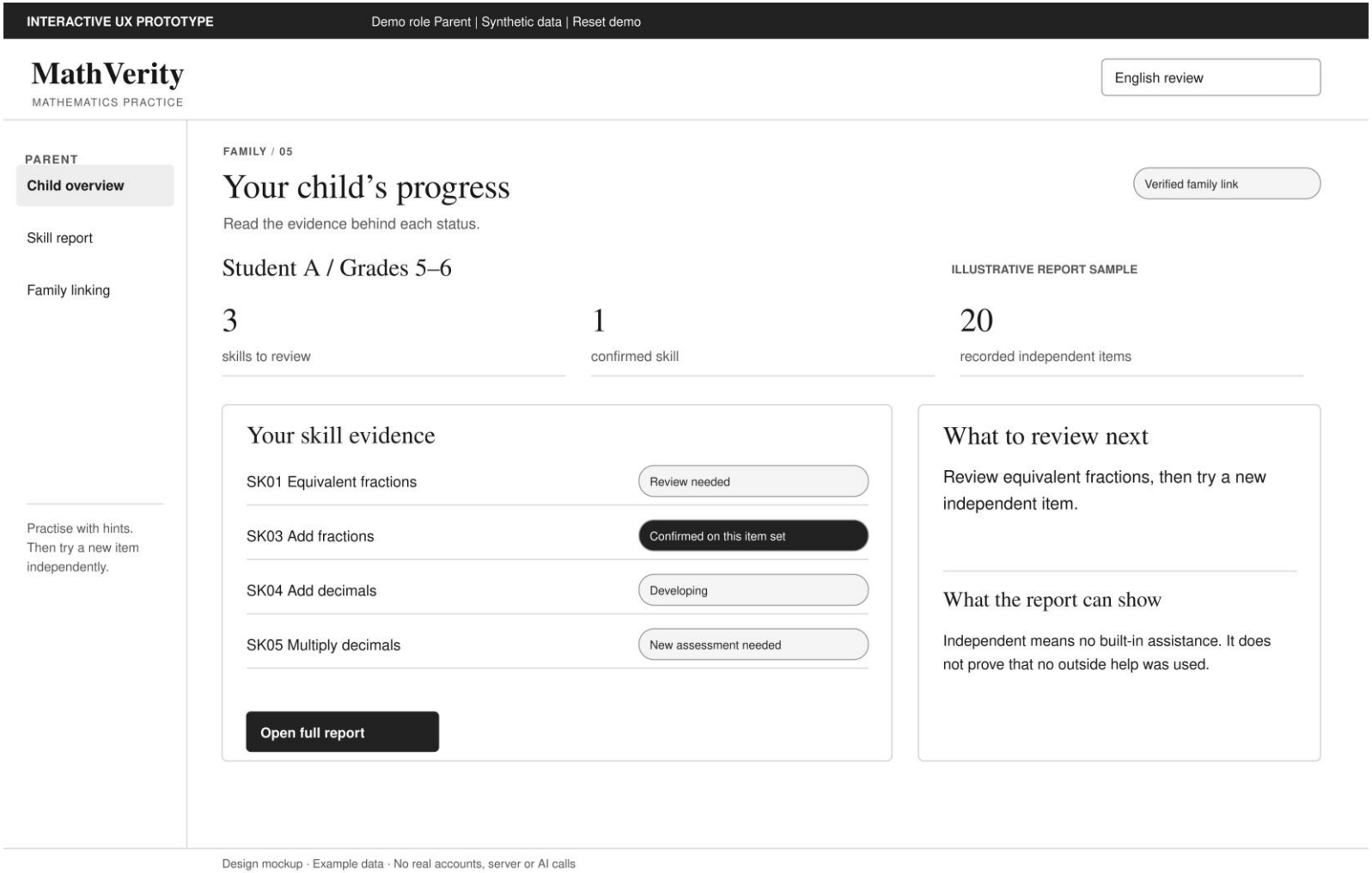


Figure 5. Linked child overview using synthetic report data.

A parent sees skills to review and an actionable recommendation. The screen states the limits of interpreting independent performance.

Parent skill report

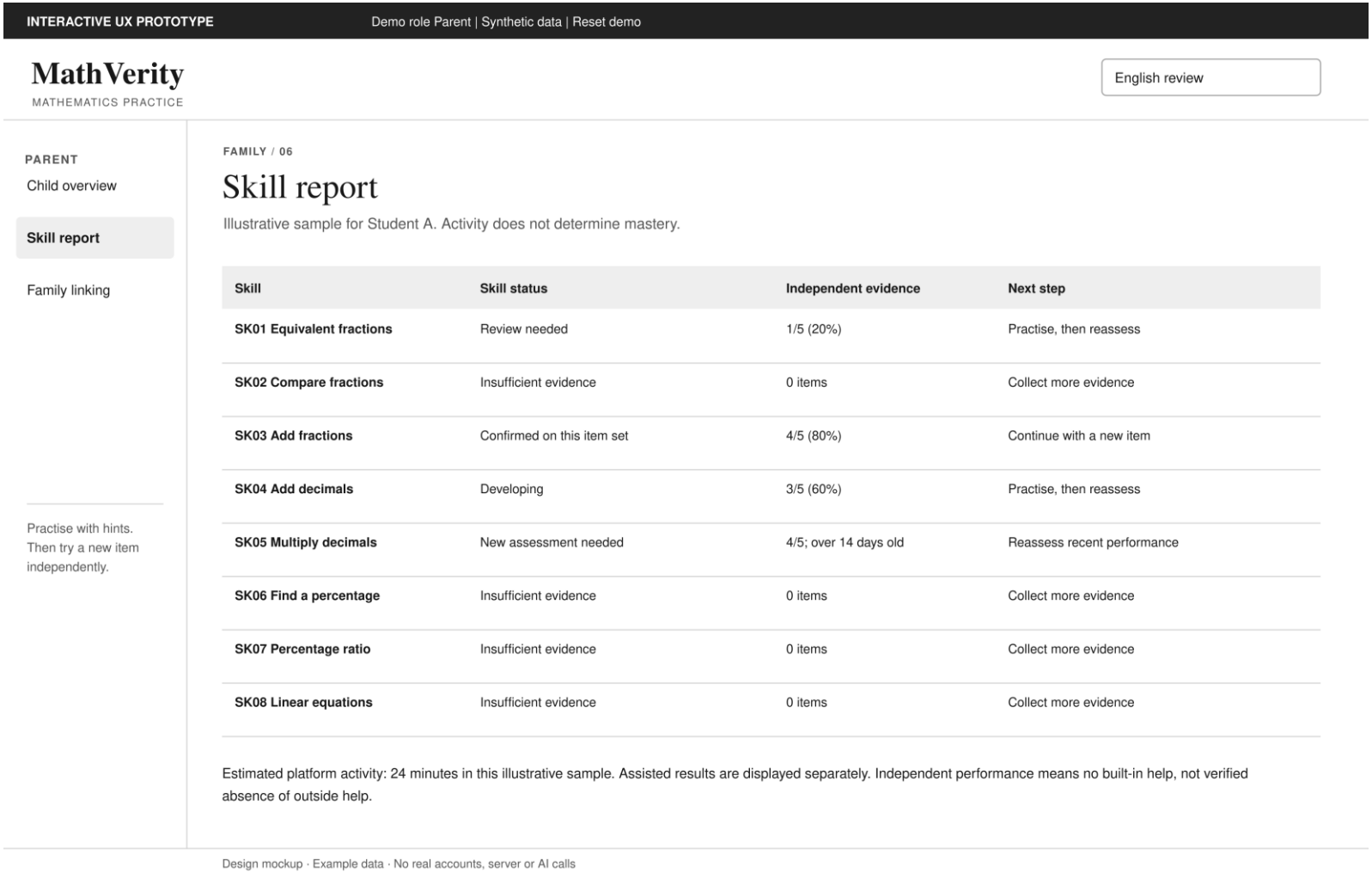


Figure 6. Skill statuses and their supporting evidence.

The report identifies the independent item count, accuracy and the next step. Old evidence requests a new assessment rather than implying current mastery.

Family linking

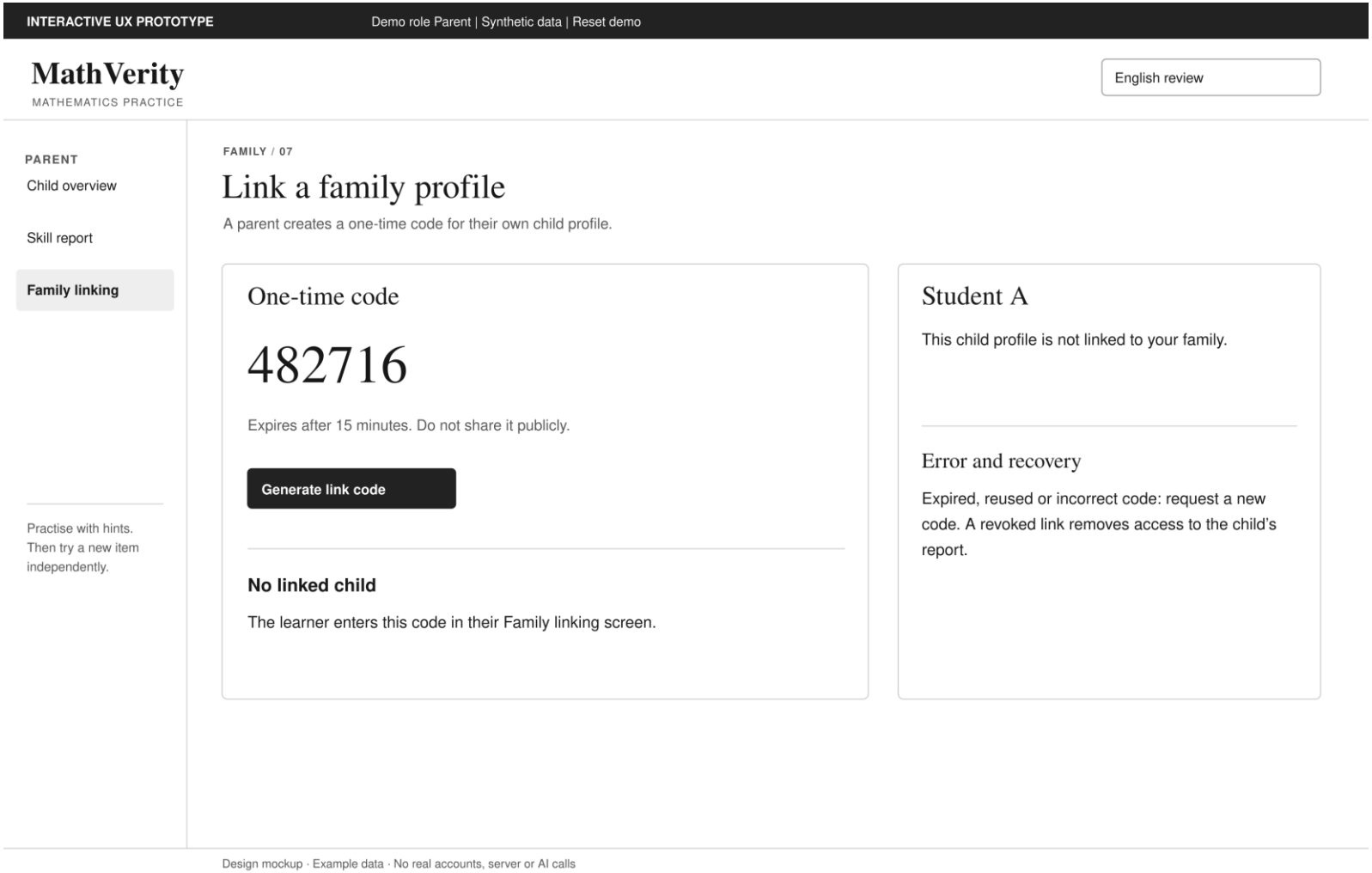


Figure 7. One-time code and recovery from linking errors.

The parent creates a code and the learner enters it. The prototype demonstrates expiry, reuse rejection and revocation; production identity verification remains pending.

Internal content review

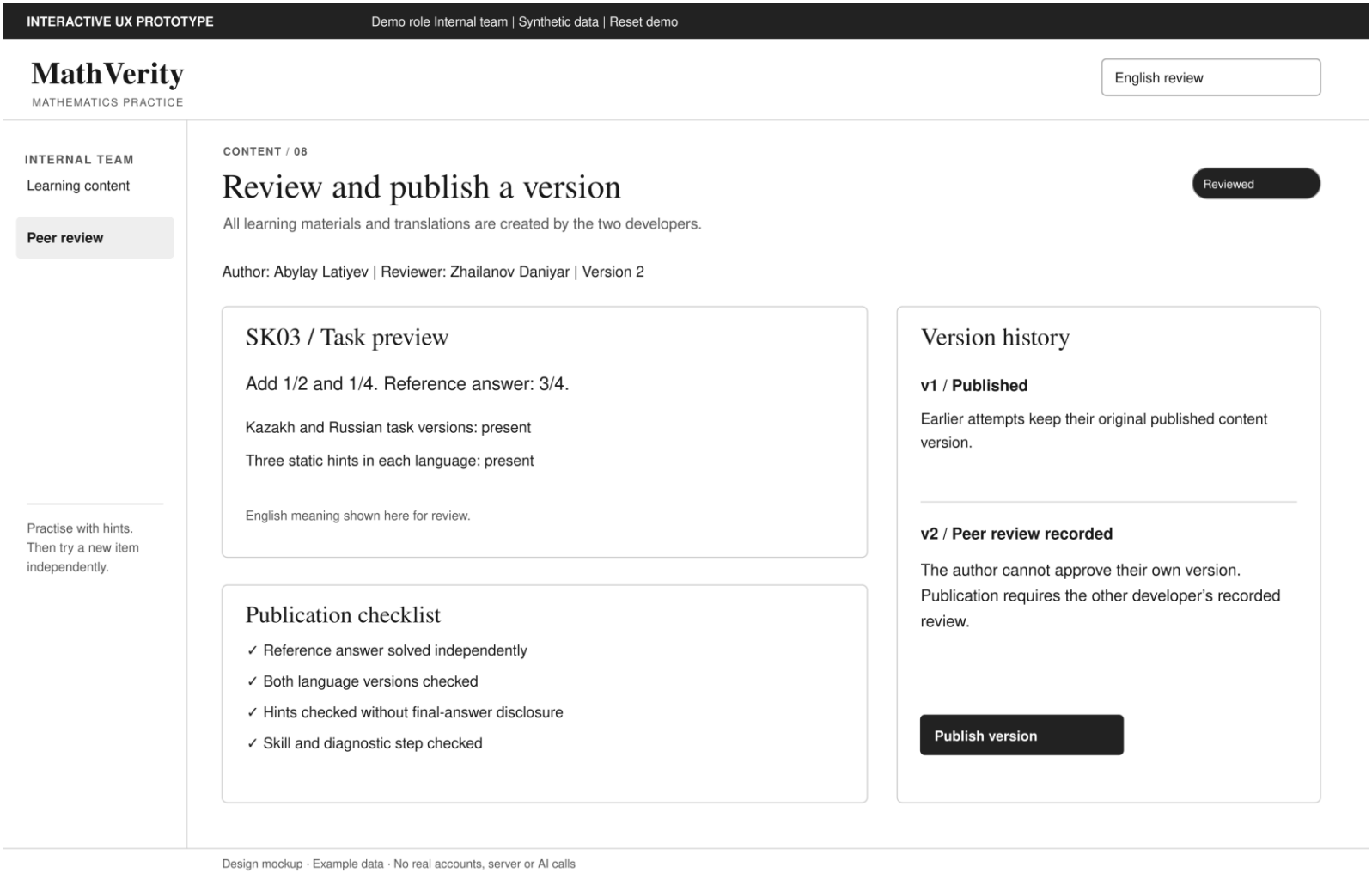


Figure 8. Peer review before publishing a bilingual content version.

The two fullstack developers create all content and translations. Publication requires a recorded review by the developer who did not author that version.

Demonstration walkthrough

The dark toolbar contains reviewer controls rather than product authentication. Use Reset demo before each walkthrough. Choose English review for the steps below; repeat the product flow in Kazakh and Russian.

Learner flow. Start practice in Fractions. Enter 2/6 and check the feedback. Request three hints, then enter 6/8; the equivalent answer is accepted. Change Hint path to Static baseline or AI simulation to inspect the source label. AI simulation uses fixed local text.

Independent flow. Open Independent check. Try 1/0: the format is rejected. Enter 10/12, submit, select the common denominator explanation and save it. Open Results: +15 XP, 1/1 correct and Insufficient evidence. Return to the item: the recorded first answer remains unchanged.

Parent flow. Select Prepared report example in Example. Open Skill report. Find SK01 review needed, SK03 confirmed, SK04 developing and SK05 requiring a new assessment. Other skills have insufficient evidence. This report is a separate synthetic fixture.

Family flow. Select Unlinked family. As Parent, open Family linking and generate a code. As Learner, enter 482716. Retry the code to see rejection. Select Expired family code to test expiry. Revoke the link and confirm that the parent report is hidden.

Team flow. Select Internal team, send version 2 for peer review and tick the four checks. Approval as Abylay is blocked. Switch Acting as to Zhailanov Daniyar, approve and publish. Create a new version to see a new draft while published history is retained.

The prototype contains four practice examples and four different independent examples. It cannot accumulate enough fresh evidence to confirm a skill through repeated play. The prepared report exists to demonstrate the other status states.

System interaction and review limits

Interaction	Demonstrated locally	Required in the production system
Answer submission	Exact fraction and decimal comparison in JavaScript.	The Go API scores supported answers and commits an immutable first attempt.
Hint request	Fixed local text, three levels and source labels.	A restricted Go adapter invokes the chosen provider; reviewed static hints remain available.
Skill report	Deterministic rule applied to synthetic records.	Authorised API queries return the linked child’s evidence and a versioned status rule.
Family link	One-time, expiry and revocation states in memory.	Verified identities, secure random codes, expiry, rate limits and object access checks.
Content publication	Team roles, review checklist and version history.	Authenticated author and reviewer identities with persisted versions and an audit trail.

Browser layout, keyboard interaction, assistive technology behaviour and narrow-screen rendering remain unverified because the preview service was unavailable. Backend security, AI quality, educational benefit and user comprehension are not established by this prototype.